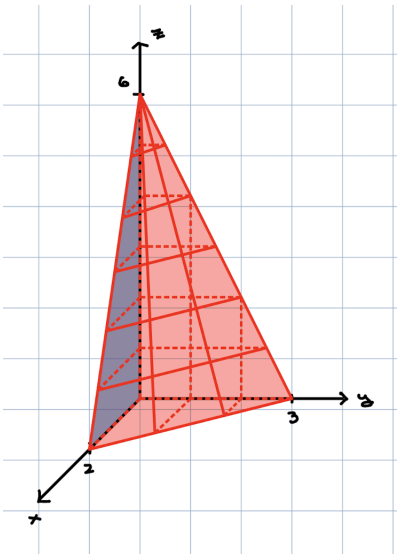
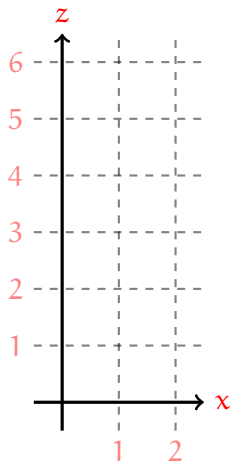
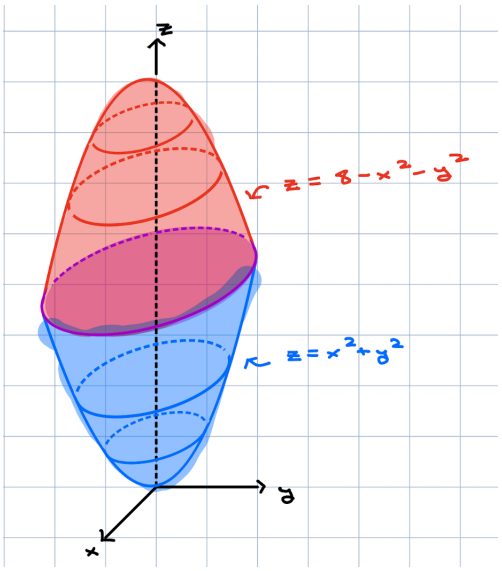
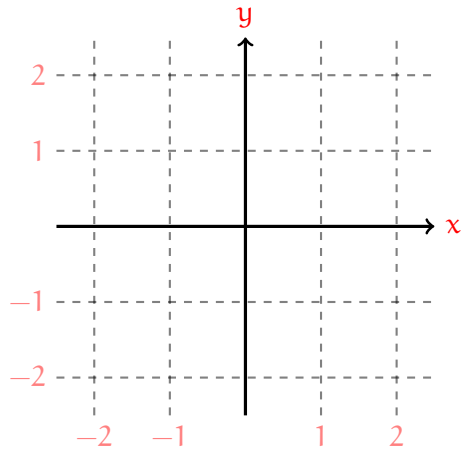
Example 1. Find the volume of the tetrahedron T bounded by the planes: $3x + 2y + z = 6$, $x = 0$, $y = 0$, and $z = 0$, but in the order: $dydzdx$. Last time, we did this in a different order of integration, and found answer 6.





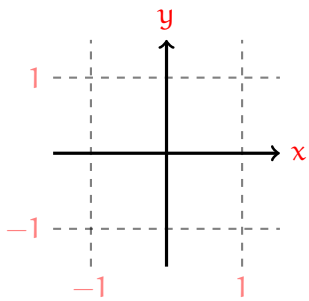
Example 2. If E is the region bounded by the paraboloids $z = 8 - x^2 - y^2$ and $z = x^2 + y^2$ then set up an iterated integral of an arbitrary function $f(x, y, z)$ over E in the in order $dzdydx$.

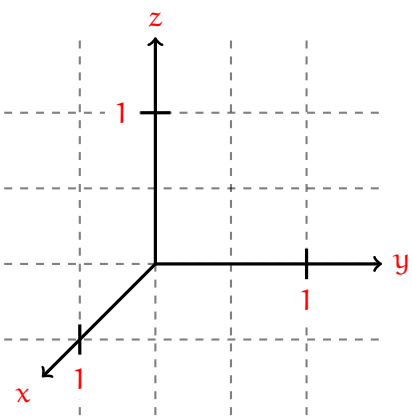




Example 3. Sketch the region of integration for:

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{x^2+y^2}} f(x,y,z)\ dzdydx$$





The equation $z = \sqrt{x^2+y^2}$ describes the top–half of a vertical double cone, i.e. describes an upwards cone, while the equation $y = \sqrt{1-x^2}$ describes the top–half of the unit circle in the xy-plane. We can also view the region in Desmos.